

Piecewise-Defined & Implicit Functions

ANSWER KEY



Evaluating piecewise functions

- 1 Given $f(x) = x^2$ for $x < 0$, and $f(x) = 2x + 1$ for $x \geq 0$, find $f(-3)$ and $f(4)$.
 $f(-3)$: since $-3 < 0$, use x^2 : $f(-3) = (-3)^2 = 9$. $f(4)$: since $4 \geq 0$, use $2x + 1$: $f(4) = 2(4) + 1 = 9$.
- 2 A shipping company charges a flat 5 dollars for packages up to 2 kg, plus 1.50 dollars per kg for anything over 2 kg. Write a piecewise function $C(w)$ for the cost, and find $C(5)$.
 $C(w) = 5$ for $0 < w \leq 2$, and $C(w) = 5 + 1.5(w - 2)$ for $w > 2$. $C(5) = 5 + 1.5(5 - 2) = 5 + 4.5 = 9.5$, so 9.5 dollars.

Checking continuity of piecewise functions

- 3 Find the value of k that makes $f(x) = 3x + k$ for $x < 2$, and $f(x) = x^2 - 1$ for $x \geq 2$, continuous at $x = 2$.
For continuity, the two pieces must agree at $x = 2$: $3(2) + k = 2^2 - 1$, so $6 + k = 3$, giving $k = -3$.
- 4 Determine whether $g(x) = x + 1$ for $x \leq 1$, and $g(x) = 3x - 2$ for $x > 1$, is continuous at $x = 1$, showing your reasoning.
Left piece at $x = 1$: $1 + 1 = 2$. Right piece as $x \rightarrow 1^+$: $3(1) - 2 = 1$. Since $2 \neq 1$, the two pieces do not match at $x = 1$, so g is not continuous there — there is a jump discontinuity.

Visualizing a piecewise graph

- 5 The graph shows $f(x) = -x - 1$ for $x < 0$, and $f(x) = x^2$ for $x \geq 0$.
 - a Evaluate the left piece and the right piece of f at $x = 0$, and use this to determine whether f is continuous there. Left piece as $x \rightarrow 0^-$: $-0 - 1 = -1$. Right piece at $x = 0$: $0^2 = 0$. Since $-1 \neq 0$, f has a jump discontinuity at $x = 0$, visible as the gap between the two branches in the graph.
 - b State the range of f shown on the graph. The linear branch for $x < 0$ takes all values greater than -1 (approaching but not including -1), and the parabola branch for $x \geq 0$ takes all values ≥ 0 . Combined range: $(-1, \infty)$.

Implicit relations and testing for functions

- 6 Determine whether $x^2 + y^2 = 25$ defines y as a function of x . Justify using the vertical line test.
This is a circle of radius 5. A vertical line such as $x = 3$ intersects it at two points ($y = 4$ and $y = -4$), so it fails the vertical line test — y is not a function of x .

- 7 Solve $x^2 + y^2 = 25$ for y explicitly, and state which branch corresponds to a function.
 $y = \pm \sqrt{25 - x^2}$. Each branch individually is a function: $y = \sqrt{25 - x^2}$ (upper semicircle) or $y = -\sqrt{25 - x^2}$ (lower semicircle) — together they form the full circle, which is not a function.
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Implicit differentiation-style reasoning (algebraic slope)

- 8 For the implicit relation $x^2 + 4y^2 = 16$ (an ellipse), find the two y -values when $x = 2$, and describe what this means about the relation as a function.
 $4 + 4y^2 = 16$, so $4y^2 = 12$, $y^2 = 3$, giving $y = \pm \sqrt{3}$. Since $x = 2$ produces two distinct y -values, the relation is not a function of x over its full domain.
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Modeling with piecewise functions

- 9 A tax schedule charges 10% on income up to 10{,}000 dollars and 20% on income above 10{,}000 dollars. Write a piecewise tax function $T(x)$ for income x , and find the tax owed on an income of 25{,}000 dollars.
 $T(x) = 0.10x$ for $0 \leq x \leq 10000$, and $T(x) = 1000 + 0.20(x - 10000)$ for $x > 10000$.
 $T(25000) = 1000 + 0.20(15000) = 1000 + 3000 = 4000$, so 4000 dollars in tax.
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Piecewise functions with three or more pieces

- 10 A parking garage charges 3 dollars for the first hour, 2 dollars for each additional hour up to 5 hours total, and a flat 15 dollars for anything over 5 hours. Write this as a three-piece function $P(h)$ for h hours parked, and find $P(4)$.
 $P(h) = 3$ for $0 < h \leq 1$; $P(h) = 3 + 2(h - 1)$ for $1 < h \leq 5$; $P(h) = 15$ for $h > 5$.
 $P(4) = 3 + 2(4 - 1) = 3 + 6 = 9$, so 9 dollars.
- 11 Evaluate the absolute value function $f(x) = |2x - 6|$ at $x = 1$ and $x = 5$ by first rewriting it as a two-piece function without absolute value bars.
Since $2x - 6 = 0$ at $x = 3$: $f(x) = -(2x - 6) = 6 - 2x$ for $x < 3$, and $f(x) = 2x - 6$ for $x \geq 3$.
 $f(1) = 6 - 2(1) = 4$. $f(5) = 2(5) - 6 = 4$.
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Graphing implicit relations point-by-point

- 12 For the implicit relation $y^2 = x + 4$, find all y -values when $x = 0$, $x = 5$, and $x = -4$, and state the smallest allowed x -value.
At $x = 0$: $y^2 = 4$, so $y = \pm 2$. At $x = 5$: $y^2 = 9$, so $y = \pm 3$. At $x = -4$: $y^2 = 0$, so $y = 0$ (a single point).
Since $y^2 \geq 0$ requires $x + 4 \geq 0$, the smallest allowed x -value is $x = -4$.
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Piecewise functions and average rate of change

- 13** For $f(x) = x^2$ on $x < 2$ and $f(x) = 3x - 2$ on $x \geq 2$, find the average rate of change of f on $[0, 4]$.

$$f(0) = 0^2 = 0 \text{ (using the first piece). } f(4) = 3(4) - 2 = 10 \text{ (using the second piece, since } 4 \geq 2\text{). AROC} \\ = \frac{10 - 0}{4 - 0} = \frac{10}{4} = 2.5.$$

- 14** Given $g(x) = 2x + 1$ for $x \leq 3$ and $g(x) = x^2 - 5$ for $x > 3$, find the average rate of change of g on $[1, 5]$.

$$g(1) = 2(1) + 1 = 3 \text{ (first piece, since } 1 \leq 3\text{). } g(5) = 5^2 - 5 = 20 \text{ (second piece, since } 5 > 3\text{). AROC} \\ = \frac{20 - 3}{5 - 1} = \frac{17}{4} = 4.25.$$

- 15** A cell phone plan charges 20 dollars flat for up to 2 GB of data, plus 8 dollars per GB for each additional GB, up to a maximum monthly charge of 68 dollars (unlimited data cap). Write a three-piece function $M(x)$ for the monthly cost based on x GB used, and find $M(8)$.

$$M(x) = 20 \text{ for } 0 \leq x \leq 2; M(x) = 20 + 8(x - 2) \text{ for } 2 < x \leq 8 \text{ (since } 20 + 8(6) = 68 \text{ is reached exactly} \\ \text{at } x = 8\text{); } M(x) = 68 \text{ for } x > 8. M(8) = 20 + 8(8 - 2) = 20 + 48 = 68, \text{ so 68 dollars.}$$